

WORK STUDY

Work study is a systematic examination of work to improve the efficient use of resources and to set standard time for performing a job.

It aims at increasing productivity, reducing cost, and improving work methods without additional investment.

Work study is the application of techniques such as **method study** and **work measurement** to analyze work and improve efficiency.

Objectives of Work Study

- To increase productivity
- To reduce unnecessary movements
- To reduce production cost
- To improve work methods
- To utilize men, machines, and materials efficiently
- To establish fair wage rates
- To improve working conditions

Components of Work Study

Work study consists of **two main techniques**:

1. Method Study

Concerned with how the work is done.

2. Work Measurement

Concerned with how long the work should take.

Method Study

Method study is the systematic recording and analysis of existing methods of doing work in order to develop an easier, safer, and more effective method.

Steps in Method Study

- Select the job
- Record the existing method
- Examine critically
- Develop improved method
- Install the new method
- Maintain the method

1. Select the Job

The first step is to select the job or operation that needs improvement.

Selection is based on:

- High cost operations
- Bottlenecks or delays
- Repetitive work
- Unsafe or fatiguing jobs

Example:

Selecting an assembly operation that consumes more time.

2. Record the Existing Method

All facts about the current method are recorded systematically using method study tools.

Tools used:

- Process charts
- Flow diagrams
- String diagrams
- Two-handed process chart

Purpose:

To get a clear and complete picture of how the work is presently done.

3. Examine the Recorded Data

The recorded method is **critically examined** to identify inefficiencies.

- What is done?
- Why is it done?
- Where is it done?
- When is it done?
- Who does it?
- How is it done?

Purpose:

To find unnecessary operations, delays, and movements.

4. Develop the Improved Method

Based on analysis, a better method is developed by:

- Eliminating unnecessary steps
- Combining operations
- Changing sequence
- Simplifying motions

- Applying principles of motion economy

Example:

Rearranging tools closer to the worker to reduce movement.

5. Install the New Method

The improved method is implemented after getting approval.

Activities include:

- Training workers
- Explaining benefits
- Updating work instructions
- Providing proper tools and layout

Purpose:

To ensure smooth adoption of the new method.

6. Maintain the New Method

The final step ensures the new method is followed continuously.

Activities:

- Regular supervision
- Periodic review
- Correcting deviations
- Updating standards if required

Purpose:

To prevent workers from returning to old methods.

Work Measurement

Work measurement determines the standard time required to complete a job by a qualified worker working at a normal pace.

Techniques of Work Measurement

- Time study
- Work sampling
- Predetermined motion time systems (PMTS)
- Standard data

Time Study:

Time study is a technique in which the time taken to perform a job is measured using a stopwatch. It is used to determine the standard time for repetitive jobs.

Procedure

- Select the job
- Break the job into elements
- Measure time for each element
- Apply performance rating
- Add allowances
- Calculate standard time

Example

Using a stopwatch to find the time taken by a worker to assemble one component on an assembly line.

Applications

- Fixing standard time
- Wage incentive plans
- Production planning

Work Sampling:

Work sampling is a statistical technique used to estimate how much time a worker or machine spends on different activities by random observations.

Key Feature

- Does not require continuous observation
- Suitable for long-cycle or irregular jobs

Example

Observing a machine operator at random intervals to find percentage of time spent on working, idle, and maintenance activities.

Applications

- Measuring machine utilization
- Studying indirect labour activities
- Maintenance work analysis

Predetermined Motion Time Systems (PMTS):

PMTS determines the time required for a job by adding standard time values for basic human motions (like reach, grasp, move), without using a stopwatch.

Common PMTS Systems

- MTM (Methods-Time Measurement)
- MOST (Maynard Operation Sequence Technique)

Example

Estimating time for assembling small electronic parts by assigning time values to motions such as reach, pick, and place.

Applications

- Short, repetitive operations
- High-precision work
- Work method comparison

Standard Data:

Meaning

Standard data refers to **pre-determined time values** collected from previous time studies for common work elements.

Key Feature

- Avoids repeated time studies
- Saves time and cost

Example

Using standard time already available for drilling a 10 mm hole instead of conducting a new time study.

Applications

- Process planning
- Cost estimation
- Production scheduling

Difference between Method Study and Work Measurement (2marks or 10 marks)

Basis	Method Study	Work Measurement
Meaning	Study of how the work is done	Study of how long the work should take
Focus	Improving work method	Fixing standard time
Main Question	<i>What is the best way to do the job?</i>	<i>How much time should the job take?</i>

Basis	Method Study	Work Measurement
Objective	Eliminate unnecessary motions and operations	Establish fair and accurate time standards
Tools Used	Process charts, flow diagrams, string diagrams, two-handed charts, motion economy	Time study, work sampling, PMTS, standard data
Outcome	Improved and simplified method	Standard time for the job
Application Stage	Done before work measurement	Done after method study
Effect on Productivity	Improves productivity by better methods	Improves productivity by time control
Nature	Qualitative analysis	Quantitative analysis
Example	Rearranging tools to reduce movement	Using stopwatch to measure operation time

Importance of Work Study

- Improves productivity
- Reduces fatigue
- Helps in manpower planning
- Fixes standard time and wages
- Improves layout and material flow
- Reduces idle time

Advantages of Work Study

- Lower production cost
- Higher output
- Better utilization of resources
- Improved safety and comfort
- Fair incentive schemes

Limitations of Work Study

- Time-consuming initially
- Requires skilled personnel
- Resistance from workers
- Not suitable for non-repetitive jobs

Example (Industry / College)

Industry: Time study used to fix standard time for assembly operation

College workshop: Method study used to improve sequence of lab operations

Technique	Main Idea	Suitable For
Time Study	Stopwatch measurement	Repetitive jobs
Work Sampling	Random observations	Long/irregular jobs
PMTS	Motion-based time values	Short manual tasks
Standard Data	Pre-established times	Repeated operations

Concept of productivity

Meaning of Productivity

Productivity refers to the efficiency with which inputs (resources) are converted into outputs (goods or services).

It shows how effectively an organization uses its men, machines, materials, money, and energy to produce results.

In simple words, productivity indicates “doing more with less.”

Definition of Productivity

Productivity is defined as the ratio of output produced to the input used in the production process.

$$\text{Productivity} = \text{Input/output}$$

- If output increases with the same input, productivity increases
- If the same output is produced with less input, productivity increases
- If output increases faster than input, productivity improves

Thus, productivity is not just about working harder, but about working smarter and more efficiently.

Example

- A worker produces 50 units/day → productivity = 50 units/day
- After method improvement, the same worker produces 70 units/day with same effort → productivity has increased

Importance of Productivity

- Reduces cost of production
- Improves competitiveness
- Increases profits
- Raises standard of living
- Improves wages and job satisfaction
- Supports economic growth

Factors Affecting Productivity

- Skill and motivation of workers
- Technology and automation
- Quality of materials
- Plant layout and work methods
- Management efficiency
- Working conditions

Process Charts, Diagrams, Work Sampling and PMTS

In work study, various analytical tools and techniques are used to study existing methods of work, measure performance, and improve productivity. Important among them are process charts, diagrams, work sampling, and predetermined motion time systems (PMTS).

1. Process Charts

Meaning

Process charts are graphical tools used to represent the sequence of activities involved in a process using standard symbols.

Activities Represented

- Operation
- Inspection
- Transportation
- Delay
- Storage

Types

- Operation Process Chart
- Flow Process Chart
- Two-Handed Process Chart

Uses

- To understand the complete process
- To identify unnecessary operations and delays

- To improve methods and workflow

Example

A process chart for manufacturing a component shows all operations from raw material to finished product, helping to eliminate waste.

2. Diagrams (Flow Diagram and String Diagram)

Meaning

Diagrams are visual representations of movement of workers, materials, or equipment in a workplace layout.

Types

- **Flow Diagram** – Shows path of movement on a layout
- **String Diagram** – Uses a string to measure distance travelled

Uses

- To analyze material and worker movement
- To reduce travel distance and fatigue
- To improve plant layout

Example

A string diagram shows a worker walking 120 meters to complete a job. After layout improvement, the distance is reduced to 40 meters.

3. Work Sampling

Meaning

Work sampling is a **statistical technique** used to estimate how time is distributed among different activities by making **random observations** over a period of time.

Features

- No continuous observation required
- Suitable for long-cycle or non-repetitive work

Uses

- Measuring machine utilization
- Studying indirect labor activities
- Analyzing idle time

Example

Observing a machine operator at random intervals to find the percentage of time spent working and idle.

4. Predetermined Motion Time Systems (PMTS)

Meaning

PMTS is a technique that determines the time required for a task by adding standard time values for basic human motions, without stopwatch timing.

Common PMTS Systems

- MTM (Methods-Time Measurement)
- MOST (Maynard Operation Sequence Technique)

Uses

- Short, repetitive manual tasks
- High-accuracy time standards
- Method comparison

Example

Calculating assembly time by adding time values for motions such as reach, grasp, and place.

Tool	Main Purpose
Process Charts	Analyze steps of a process
Diagrams	Study movement and layout
Work Sampling	Measure time distribution
PMTS	Predict time from motions

Principles of Motion Economy

Principles of Motion Economy are a set of guidelines used in method study to design work in such a way that unnecessary motions are eliminated, worker fatigue is reduced, and productivity is increased.

These principles help in achieving maximum output with minimum effort by improving the use of human motions, workplace arrangement, and tools/equipment.

Motion economy refers to the efficient use of human motions during work so that tasks are performed with least physical effort, minimum time, and reduced fatigue.

Classification of Principles of Motion Economy

The principles are generally classified into three groups:

1. Principles related to use of the human body
2. Principles related to arrangement of the workplace
3. Principles related to design of tools and equipment

Principles Related to Use of the Human Body

1. **Both hands should work simultaneously**

Idle time of either hand should be avoided.

2. **Hands should not be idle at the same time**

When one hand is waiting, the other should be doing useful work.

3. **Use symmetrical and natural motions**

Movements of both hands should be balanced and coordinated.

4. **Use minimum number of motions**

Eliminate unnecessary reaching, bending, and twisting.

5. **Prefer smooth, continuous motions**

Jerky and sudden movements increase fatigue.

6. **Use momentum wherever possible**

Momentum reduces muscular effort.

Principles Related to Workplace Arrangement

1. **A fixed place for tools and materials**

Reduces search time.

2. **Tools and materials should be within normal reach**

Avoid excessive stretching and bending.

3. **Materials should be arranged in sequence of use**

Saves time and confusion.

4. **Use gravity feed and drop delivery**

Reduces manual handling effort.

5. Proper working height

Work should be performed at waist or elbow height to reduce strain.

Principles Related to Design of Tools and Equipment

1. Use jigs and fixtures

Reduces manual effort and improves accuracy.

2. Combine tools where possible

Reduces tool changes.

3. Tools should be easy to handle

Proper grip and weight reduce fatigue.

4. Mechanize repetitive tasks

Reduces human effort and increases speed.

5. Use foot-operated devices

Frees hands for productive work.

Example (Industrial / Workshop)

In an assembly operation:

- Tools are placed within easy reach
- Both hands are used simultaneously
- Gravity-fed bins are used for components

This reduces operation time and worker fatigue.

Advantages of Applying Motion Economy Principles

- Increased productivity
- Reduced worker fatigue
- Improved safety
- Better work quality
- Reduced operation time
- Efficient use of manpower

Applications

- Assembly lines
- Packing operations
- Workshop practices
- Manufacturing and service industries

Micro motion study

Micro motion study is an advanced technique of method study used to analyze very small, detailed motions performed by a worker while doing a job.

It is mainly used for highly repetitive, short-cycle manual operations where even small time savings are important.

Micro motion study is the technique of recording and analyzing the basic human motions (therbligs) of a worker using motion picture cameras or video recording to improve work methods and reduce unnecessary movements.

Objectives of Micro Motion Study

- To eliminate unnecessary motions
- To reduce operation time
- To reduce worker fatigue
- To improve productivity
- To develop the best method of work
- To standardize efficient work methods

When Micro Motion Study is used

- Short cycle, repetitive operations
- Assembly and packing jobs
- Precision work
- High production volume situations

Basic Concept – Therbligs

Micro motion study is based on **therbligs**, which are the **basic elements of human motion**.

Examples of therbligs:

- Reach
- Grasp
- Move
- Hold
- Release
- Search
- Select
- Inspect

Each therblig is analyzed to see whether it is necessary or can be eliminated.

Procedure of Micro Motion Study

- Select the job to be studied
- Record the activity using a high-speed camera or video
- Analyze the film frame by frame
- Identify therbligs involved
- Eliminate unnecessary motions
- Develop improved method
- Implement and standardize the method

Recording Tools Used

- High-speed video camera
- Micro motion camera
- SIMO chart (Simultaneous Motion Chart)
- Frame-by-frame analysis software

Example

Industrial Example

In an electronic assembly job, micro motion study reveals excessive hand movements during component picking. By rearranging component bins closer to the worker, unnecessary reach and search motions are eliminated, reducing cycle time.

Workshop Example

In a college workshop assembly task, micro motion study helps students learn correct hand movements, reducing fatigue and improving speed.

Advantages of Micro Motion Study

- Highly accurate analysis
- Detects even very small inefficiencies
- Reduces cycle time significantly
- Improves work quality
- Reduces fatigue
- Helps in training workers

Limitations of Micro Motion Study

- Expensive equipment required
- Time-consuming analysis
- Not suitable for long or non-repetitive jobs
- Requires skilled analysts

Difference between Micro Motion Study and Time Study

Basis	Micro Motion Study	Time Study
Meaning	Detailed study of very small human motions	Measurement of time taken to perform a job
Purpose	Improve work method by eliminating unnecessary motions	Determine standard time for a job
Level of Detail	Very high (minute motions)	Moderate (job elements)
Recording Technique	Motion picture / video recording	Stopwatch timing
Basic Unit Studied	Therbligs (reach, grasp, move, etc.)	Work elements or operations
Main Focus	Method improvement	Time measurement
Accuracy	Very high	Good
Cost	High (special equipment)	Low (simple tools)
Time Required	Time-consuming analysis	Faster to conduct
Suitable For	Short-cycle, repetitive manual work	Repetitive production jobs
Output	Best method of work	Standard time
Worker Fatigue Analysis	Excellent	Limited

Therbligs

Therbligs are the basic elements of human motion used in micro motion study to analyze work at a very detailed level.

The term *Therblig* is derived from the name Gilbreth introduced by Frank and Lillian Gilbreth.

Each therblig represents a small, identifiable motion made by a worker while performing a task.

Therbligs are the fundamental motions of the human body used to study, analyze, and improve manual work by eliminating unnecessary movements.

Purpose of Therbligs

- To study work at motion level
- To identify inefficient or unnecessary motions
- To reduce time and fatigue
- To develop the best work method
- To improve productivity and efficiency

List of Basic Therbligs

Therblig	Meaning
Search (Sh)	Looking for an object
Find (F)	Locating the object
Select (Se)	Choosing one item from many
Grasp (G)	Taking hold of an object
Hold (H)	Holding an object
Reach (R)	Moving hand to an object
Move (M)	Moving object to another location
Release Load (RL)	Letting go of the object
Position (P)	Aligning or placing an object
Use (U)	Using a tool or object
Assemble (A)	Joining two parts
Disassemble (DA)	Separating parts
Inspect (I)	Checking quality or condition
Pre-position (PP)	Placing object ready for next step
Transport Loaded (TL)	Moving with load
Transport Empty (TE)	Moving without load
Delay (D)	Waiting (avoidable or unavoidable)
Rest (R)	Recovery from fatigue
Plan (Pn)	Thinking or decision-making

Classification of Therbligs

A) Effective (Productive) Therbligs

These contribute directly to work:

- Grasp
- Move
- Use
- Assemble
- Release load

B) Ineffective (Non-productive) Therbligs

These should be minimized or eliminated:

- Search
- Find
- Delay
- Hold
- Rest (excessive)

Application of Therbligs

Therbligs are mainly used in:

- Micro motion study
- SIMO charts (Simultaneous Motion Charts)
- Assembly and packing operations
- Short-cycle repetitive jobs

Example

In an assembly operation:

- **Search** and **find** time is reduced by keeping tools at a fixed location
- **Reach** and **move** distances are minimized
- **Assemble** and **use** motions are optimized

This reduces cycle time and fatigue.

Advantages of Using Therbligs

- Highly detailed motion analysis
- Helps eliminate wasteful motions
- Reduces fatigue
- Improves productivity
- Useful for training workers

Limitations

- Requires special recording equipment
- Time-consuming analysis
- Not suitable for non-repetitive jobs

SIMO chart (Simultaneous Motion Chart)

A SIMO chart (Simultaneous Motion Chart) is a micro motion study tool used to record and analyze the simultaneous movements of both hands (or different body parts) of a worker while performing a task.

A SIMO chart is a chart that records the therbligs (basic motions) of the left hand and right hand simultaneously on a common time scale.

Purpose of SIMO Chart

- To study minute human motions
- To ensure both hands work effectively
- To eliminate unnecessary or idle motions
- To reduce fatigue
- To improve productivity and efficiency

Basis of SIMO Chart

SIMO charts are based on therbligs, which are the basic elements of human motion such as:

- Reach
- Grasp
- Move
- Hold
- Release
- Assemble
- Inspect

When SIMO Chart is used

- Very short cycle operations
- Highly repetitive assembly work
- Precision work
- Where small time savings are important

Procedure for Preparing a SIMO Chart

- Select the operation to be studied
- Record the operation using a video or motion picture camera
- Analyze the recording frame by frame
- Identify therbligs for left hand and right hand
- Record the therbligs on a time scale
- Analyze and improve the method

Structure of a SIMO Chart

- Two horizontal rows:
 - One for **Left Hand**
 - One for **Right Hand**
- Time is represented on the horizontal axis
- Each motion (therblig) is shown in sequence

Example:

Assembly Example

In a small component assembly:

- SIMO chart shows the left hand idle while the right hand works
- Method is improved by rearranging parts so both hands work simultaneously

Result: Reduced cycle time and fatigue.

Advantages of SIMO Chart

- Very detailed motion analysis
- Helps balance hand movements
- Reduces unnecessary motions
- Improves work methods
- Useful for training workers

Limitations of SIMO Chart

- Requires special recording equipment
- Time-consuming and costly
- Not suitable for long-cycle or non-repetitive jobs
- Requires skilled analysts

SIMO chart is a powerful tool of micro motion study that helps analyze and improve very small human motions. Though expensive and time-consuming, it is highly effective for short, repetitive operations where even small improvements lead to significant productivity gains.

Difference between Two-Handed Process Chart and SIMO Chart (2 marks)

Two-Handed Chart	SIMO Chart
Records hand activities	Records therbligs
Less detailed	Highly detailed
Manual observation	Camera-based analysis

Work measurement – Stop watch procedure of time study

Time study is one of the most important techniques of work measurement. It is used to determine the standard time required by a qualified worker to perform a task at a normal pace, using a stopwatch.

The stopwatch procedure of time study helps management in planning, scheduling, costing, wage fixation, and productivity improvement.

Stopwatch time study is a technique in which the time taken to perform each element of a job is measured using a stopwatch, and then analyzed to establish the standard time.

Objectives of Stopwatch Time Study

- To determine standard time for an operation
- To improve productivity
- To plan manpower and machines
- To fix fair wages and incentive schemes
- To estimate production cost
- To compare work methods

Equipment Used

- Stopwatch (decimal or digital)
- Time study board
- Time study sheet
- Pencil / pen

Stopwatch Procedure of Time Study

Step 1: Selection of Job

The first step is to select the job or operation to be studied.

The job selected should be:

- Repetitive
- Important in terms of cost or time
- Bottleneck operation

Example:

Selecting a drilling operation that consumes more time.

Step 2: Selection of Worker

A qualified and experienced worker performing the job at a normal pace is selected.

The worker should:

- Be familiar with the job
- Follow the standard method
- Not be exceptionally fast or slow

Step 3: Breakdown of Job into Elements

The job is divided into small elements for easy observation and accurate timing.

Each element should:

- Have a clear start and end
- Be repetitive
- Take a measurable time

Example:

- Pick the component
- Position the component
- Drill the hole
- Remove the component

Step 4: Choose Timing Method

Two methods are used:

1. **Continuous Timing Method**
 - Stopwatch runs continuously
 - Time readings are taken at the end of each element
2. **Flyback (Snapback) Timing Method**
 - Stopwatch is reset to zero after each element
 - Time for each element is recorded directly

Step 5: Record Observed Time

The analyst observes the worker and records the time for each element using the stopwatch.

- Several cycles are observed
- More observations improve accuracy

Step 6: Calculate Average Observed Time

For each element, the average observed time is calculated from all recorded cycles. This reduces the effect of abnormal variations.

Step 7: Performance Rating

Performance rating adjusts the observed time based on the speed of the worker relative to normal performance.

- If worker is faster → rating > 100%
- If worker is slower → rating < 100%

Formula:

Normal Time = Observed Time × Performance Rating

Step 8: Add Allowances

Allowances are added to compensate for unavoidable delays.

Types of allowances:

- Personal needs allowance
- Fatigue allowance
- Delay allowance

Step 9: Calculate Standard Time

Standard time is the time allowed to complete the job under normal working conditions.

Formula:

Standard Time=Normal Time + Allowances

Advantages of Stopwatch Time Study

- Simple and easy to use
- Accurate for repetitive jobs
- Widely accepted technique
- Helps in cost control and planning
- Useful for wage fixation

Limitations of Stopwatch Time Study

- Time-consuming
- Causes worker nervousness
- Not suitable for non-repetitive jobs
- Depends on skill of time study analyst

Applications

- Assembly lines
- Machining operations
- Workshop practices
- Manufacturing industries

Example (Workshop / Industry)

In a college workshop, stopwatch time study is used to determine the standard time for drilling a hole on a component by measuring the time for each element and adding allowances.

Performance rating

Performance rating is the process of assessing the speed and effectiveness of a worker while performing a job, relative to a standard or normal pace.

It is used in time study to convert observed time into normal time.

Performance rating is the adjustment factor applied to the observed time to account for the worker's speed, skill, and effort compared to standard performance.

Purpose of Performance Rating

- To eliminate the effect of worker speed variation
- To determine normal time
- To ensure fair and accurate time standards
- To standardize time measurement irrespective of the worker

Concept of Normal Performance

- Normal performance = the pace at which a trained worker works comfortably and steadily without undue strain
- Represented as 100% rating

If:

- Worker is faster → Rating > 100%
- Worker is slower → Rating < 100%

Formula Used

- Normal Time = Observed Time × Performance Rating

Example

Observed time for an operation = **5 minutes**

Worker is judged to be working at **120% speed**

Normal Time = $5 \times 1.20 = 6$ minutes

Methods of Performance Rating

1. Speed Rating

- Based only on worker's speed
- Simple but less accurate

2. Westinghouse System

Considers four factors:

- Skill
- Effort
- Conditions

- Consistency

More accurate and widely used.

3. Synthetic Rating

Uses predetermined standard times for elements.

4. Objective Rating

Separates **speed** and **job difficulty**.

Importance of Performance Rating

- Ensures accurate standard time
- Prevents bias due to fast or slow workers
- Helps in fair wage fixation
- Improves planning and scheduling accuracy
- Supports productivity measurement

Applications

- Time study
- Work measurement
- Incentive wage plans
- Production planning and control

Allowances (in Time Study / Work Measurement)

Allowances are extra time added to the normal time to account for unavoidable delays, fatigue, and personal needs of a worker.

They ensure that the standard time is fair, realistic, and achievable under normal working conditions.

Allowances are defined as the additional time provided over normal time to compensate for personal needs, fatigue, and unavoidable delays while performing a job.

Need for Allowances

Allowances are necessary because:

- Workers are not machines
- Continuous work causes fatigue
- Small interruptions are unavoidable
- Workers need time for personal requirements

Without allowances, standard time would be too tight and unfair.

Relationship Between Times

Standard Time=Normal Time + Allowances

Or

Standard Time=Normal Time $\times$ (1+Allowance factor)

Types of Allowances

1. Personal Allowance

Personal allowance is given to meet personal needs of workers.

Examples

- Drinking water
- Washroom use
- Personal comfort

Typical Value

- Usually 5% of normal time

2. Fatigue Allowance

Fatigue allowance compensates for physical and mental tiredness caused due to work.

Factors Affecting Fatigue

- Nature of work (light/heavy)
- Working posture
- Temperature and environment
- Repetitiveness of job

Typical Value

- Ranges from 4% to 10% or more depending on job severity

3. Delay Allowance

Delay allowance is given for unavoidable interruptions during work.

Examples

- Tool adjustments
- Machine breakdowns
- Waiting for material
- Power failure

Types

- Unavoidable delay allowance – allowed
- Avoidable delay allowance – not encouraged

Typical Value

- Around 3% to 5%

4. Contingency Allowance

Given for occasional or irregular activities that do not occur every cycle.

Examples

- Cleaning machine once per shift
- Minor inspection work

Typical Value

- Usually less than 5%

5. Special Allowances

These are provided for special conditions.

Examples

- Policy allowance (management decision)
- Learning allowance (new workers)
- Setup allowance
- Tool change allowance

Importance of Allowances

- Ensures fair standard time
- Reduces worker fatigue
- Improves job satisfaction
- Prevents over-pressure on workers
- Helps in realistic planning
- Improves productivity in the long run

Consequences of Improper Allowances

- Too low → worker dissatisfaction, fatigue
- Too high → loss of productivity, higher cost

Hence, allowances must be carefully decided.

Applications of Allowances

- Time study
- Standard time calculation
- Incentive wage plans
- Production planning and scheduling